

A FORMALIZATION OF THE ELLIGATOR FAMILY OF UNIFORM ENCODINGS TO ELLIPTIC CURVES

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ABSTRACT. First Elligator formalization (what/who), What formalized (Ell1/2, Legendre), where stored (CSLib, what?), advances (prime power, computability, test vector testbenching...)

1. INTRODUCTION

Elliptic-curve cryptography routinely transmits points on a fixed public curve, and such points are easily distinguished from uniform random bit strings: an observer who can tell elliptic-curve traffic apart from random traffic can selectively block it. Bernstein, Hamburg, Krasnova and Lange’s Elligator [BHKL13] addresses this by giving efficient, invertible encodings between (most of) the points of a suitable elliptic curve and uniform bit strings, so that a random curve point can be transmitted as what looks like a random string. The paper introduces two such encodings, Elligator 1 and Elligator 2, together with Curve1174, a concrete Edwards curve suited to Elligator 1. Later work has extended the approach to further curve shapes, including a Legendre-curve variant due to Saha and Karati [SK26].

Despite substantial practical uptake — Elligator has been numerically tested [BHKL] and implemented across a range of libraries and protocols [ell] — the underlying theory has, to our knowledge, never been formalized in a proof assistant. This paper reports on such a formalization, carried out in CSLib, a Lean 4 library that currently has essentially no cryptographic primitives, not even basic elliptic-curve definitions. Elligator is accordingly the first substantial elliptic-curve encoding family to enter CSLib.

Contributions.

- (1) A formalization of elliptic curves and the curve primitives needed to state Elligator, none of which previously existed in CSLib.
- (2) A formalization of Elligator 1 and Curve1174, and a generalization of the viability lemmas underlying Elligator 1 from prime fields to prime power fields (Section 3).
- (3) A formalization of Elligator 2 and of the Legendre-curve variant of Saha and Karati as instances of a common encoding family alongside Elligator 1 (Section 4).

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- (4) Computable, extractable definitions throughout, so that the formalized encodings are not merely existence statements but can be run and checked against published test vectors (Section 6).

Outline. Section 2 recalls the curve and field background. Section 3 covers Elligator 1 and its generalization. Section 4 covers Elligator 2 and the Legendre-curve variant. Section 5 covers Curve1174. Section 6 discusses computability and validation against test vectors. Section 7 discusses related work and Section 8 concludes.

2. BACKGROUND

This section fixes the curve and field notions used throughout, all newly formalized in CSLib for this work unless noted otherwise.

state the actual CSLib definitions of finite fields, quadratic residues/the Legendre symbol (or point at Mathlib’s, if reused), and the curve shapes used below (Edwards, Montgomery, Legendre), matching the Lean source exactly.

2.1. Finite fields and quadratic residues.

2.2. Curve shapes. definitions for the (twisted) Edwards curves used by Elligator 1, the Montgomery curves used by Elligator 2, and the Legendre curves $E_\lambda : y^2 = x(x-1)(x-\lambda)$ used by the Legendre-curve variant.

3. ELLIGATOR 1 AND ITS GENERALIZATION TO PRIME POWERS

state Bernstein et al.’s original viability conditions and lemmas, then generalized statements side by side, and give a proof sketch of the generalization.

Elligator 1 [BHK13, §3] encodes bit strings as points on a complete Edwards curve $x^2 + y^2 = 1 + dx^2y^2$ over a prime field $\mathbb{F}_p$ with $p \equiv 3 \pmod{4}$, for d of a specific non-square form. The viability of the construction — that the encoding is well-defined and injective on the intended domain — is established in the original paper for prime p .

3.1. Generalizing to prime powers. We show that the viability lemmas underlying Elligator 1 do not in fact require p to be prime, only that the underlying field has $q = p^k$ elements with $q \equiv 3 \pmod{4}$: that is, the construction is viable over any finite field of order a prime power congruent to 3 mod 4, not only prime fields. This is, to our knowledge, a genuine extension of Bernstein et al.’s result rather than a restatement of it.

Theorem 3.1 (Generalized Elligator 1 viability). *TODO placeholder Let $q = p^k$ be a prime power with $q \equiv 3 \pmod{4}$. Then the Elligator 1 encoding is well-defined and injective on [TODO: the intended domain] over $\mathbb{F}_q$.*

4. ELLIGATOR 2 AND THE LEGENDRE-CURVE VARIANT

We formalize both of these alongside Elligator 1, collecting the three under a common presentation in CSLib.

4.1. **Elligator 2.** TODO formal statement and theorem.

4.2. **The Legendre-curve variant.** TODO formal statement and theorem.

4.3. **A common family.** TODO if there is a genuine shared interface/theorem here (e.g. a common correctness criterion all three encodings satisfy), state it. If not, at minimum give a comparison table of domain, codomain, applicable curve shapes, and injectivity radius across the three encodings.

5. CURVE1174

TODO state the concrete curve equation/parameters, and which properties of it (completeness, group order, twist security, suitability for Elligator 1) are formalized vs. assumed/imported.

Curve1174 [BHK13, §4] is the concrete Edwards curve $x^2 + y^2 = 1 - 1174x^2y^2$ over $\mathbb{F}_p$ with $p = 2^{251} - 9$, introduced alongside Elligator 1 as a curve satisfying the conditions required for the encoding together with the security properties expected of a cryptographic curve. We formalize its definition and the properties needed to instantiate the Elligator 1 encoding of Section 3 on it.

6. COMPUTABILITY AND VALIDATION

TODO describe the extraction/execution mechanism used (Lean’s eval, code generation, or otherwise), and report the actual validation.

A definition that merely proves the existence of an encoding is of limited practical use; we give computable, extractable definitions for every encoding in this paper, so that they produce actual byte sequences that can be compared against independent implementations. This matters for CSLib’s stated goal of being practically reusable, not only mathematically faithful.

6.1. **Extraction.** TODO how the Lean definitions are made executable, and any performance caveats.

6.2. **Validation against published test vectors.** TODO report which test vectors were checked (e.g. from elligator.cr.yp.to or elligator.org) and the outcome.

We cross-checked the extracted implementations of Section 6.1 against the test vectors published alongside the original Elligator paper [BHK13] and the reference implementations catalogued at [ell].

7. RELATED WORK

The Elligator constructions have been extended and adapted in several directions we do not formalize here, including Elligator Squared [Tib14], which supports curves of prime order not covered by the original constructions, and Elligator-T [SK26] for twisted Edwards curves. To our knowledge, none of the Elligator family has previously been formalized in a proof assistant:

existing validation has been numerical [BHKL] or by way of independent implementations [ell], rather than machine-checked proof.

CSLib itself has, prior to this work, essentially no cryptographic primitives; a separate, unrelated CSLib effort formalizes binding representations for the untyped lambda calculus, sharing this paper’s goal of building reusable, well-reviewed foundations rather than one-off developments, but is otherwise unconnected in subject matter.

8. CONCLUSION

We have given the first formalization of the Elligator family of uniform encodings to elliptic curves, covering Elligator 1 (with a generalization of its viability lemmas from prime to prime power fields), Elligator 2, the Legendre-curve variant, and Curve1174, together with computable definitions validated against published test vectors. This is CSLib’s first substantial cryptographic development, and we hope it serves as a starting point for further elliptic-curve formalization within the library.

REFERENCES

- [BHKL] Daniel J. Bernstein, Mike Hamburg, Anna Krasnova, and Tanja Lange. Elligator: numerical tests and papers. <https://elligator.cr.yp.to/papers.html>. Accessed 2026.
- [BHKL13] Daniel J. Bernstein, Mike Hamburg, Anna Krasnova, and Tanja Lange. Elligator: elliptic-curve points indistinguishable from uniform random strings. In *Proceedings of the 2013 ACM SIGSAC Conference on Computer & Communications Security, CCS '13*, page 967–980, New York, NY, USA, 2013. Association for Computing Machinery.
- [ell] Elligator: implementations. <https://elligator.org/implementations>. Accessed 2026.
- [SK26] Gourab Chandra Saha and Sabyasachi Karati. Encoding to legendre curve, kummer line and twisted edwards curve. *Advances in Mathematics of Communications*, 23(0):52–90, 2026.
- [Tib14] Mehdi Tibouchi. Elligator squared: Uniform points on elliptic curves of prime order as uniform random strings. In Nicolas Christin and Reihaneh Safavi-Naini, editors, *Financial Cryptography and Data Security - 18th International Conference, FC 2014, Christ Church, Barbados, March 3-7, 2014, Revised Selected Papers*, volume 8437 of *Lecture Notes in Computer Science*, pages 139–156. Springer, 2014.

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